



Centre for Marine Science and Technology

**Analysis of Low Frequency Underwater Acoustic Signals
Possibly Related to the Loss of Malaysian Airlines Flight
MH370**

Prepared for:
Australian Transport Safety Bureau

Prepared by:
Dr Alec J Duncan, Dr Alexander Gavrilov, Dr Rob McCauley

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Executive Summary

Malaysian Airlines flight MH370 disappeared in the early hours of 8th March 2014 and its whereabouts remain unknown. Current search efforts are based on position information derived from a satellite communication system and are focussed on the Indian Ocean, within Australia's search and rescue zone. The Australian Transport Safety Bureau (ATSB) has asked the Centre for Marine Science and Technology (CMST) to analyse signals received on underwater sound recorders operated by CMST that form part of the Australian Government funded Integrated Marine Observing System (IMOS), and on hydroacoustic stations operated by the Comprehensive Nuclear Test Ban Treaty Organisation (CTBTO) in an attempt to detect and localise underwater sounds that could be associated with the impact of the aircraft on the water or with the implosion of wreckage as the aircraft sank.

One acoustic event of particular interest has been identified that occurred at a time that could potentially link it to MH370 and appears to have been received on one of the IMOS recorders near the Perth Canyon (RCS) and at the CTBTO hydroacoustic station at Cape Leeuwin (HA01).

A detailed analysis of these signals has resulted in an approximate localisation for the source that is compatible with the time of the last satellite handshake with the aircraft, but incompatible with the satellite to aircraft range derived from this handshake. There appear to be three possible explanations for this discrepancy:

1. The signals received at HA01 and RCS are from the same acoustic event, but the source of the signals is unrelated to MH370.
2. The signals received at HA01 and RCS are from different acoustic events, which may or may not be related to MH370.
3. The signals received at HA01 and RCS are from the same acoustic event, and the source of the signals is related to MH370, but there is a problem with the position line determined from the satellite handshake data.

Of these, the first explanation seems the most likely as the characteristics of the signals are not unusual, it is only their arrival time and to some extent the direction from which they came that make them of interest.

If the second explanation was correct then there would still be some prospect that the signal received at HA01 could be related to the aircraft, in which case the combination of the HA01 bearing and the position arc derived from the satellite handshake data would provide an accurate location on which to base a search. However, the analysis carried out here indicates that, while not impossible, this explanation is unlikely.

The third explanation also seems unlikely because of the intense scrutiny the satellite handshake data has been subjected to. However, should the arc defined by the handshake data be called into question, the various timing and acoustic considerations discussed here would suggest that a reasonable place to look for the aircraft would be near where the position line defined by a bearing of 301.6° from HA01 crosses the Chagos-Laccadive Ridge, at approximately 2.3°S , 73.7°E . If the source of the detected signals was the aircraft impacting the sea surface then this would most likely have occurred in water depths less than 2000m and where the seabed slopes downwards towards the east or southeast. These considerations could be used to further refine the search area. If, instead, the received sounds were due to debris imploding at depth it is much less certain where along the position line from HA01 this would have occurred.

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1 Introduction

Malaysian Airlines flight MH370 disappeared in the early hours of 8th March 2014 and its whereabouts remain unknown. Current search efforts are based on position information derived from a satellite communication system and are focussed on the Indian Ocean, within Australia's search and rescue zone. The Australian Transport Safety Bureau (ATSB) has asked the Centre for Marine Science and Technology (CMST) to analyse signals received on underwater sound recorders operated by CMST as part of the Integrated Marine Observing System (IMOS) and on hydroacoustic stations operated by the Comprehensive Nuclear Test Ban Treaty Organisation (CTBTO) in an attempt to detect and localise underwater sounds that could be associated with the impact of the aircraft on the water or with the implosion of wreckage as the aircraft sank. This report collects together the results of various analyses that have been carried out by CMST, several of which have previously been reported to ATSB in summary reports.

One acoustic event of particular interest has been identified that occurred at a time that could potentially link it to MH370 and appears to have been received on CMST's acoustic recorders near the Perth Canyon (RCS), and at the CTBTO hydroacoustic station at Cape Leeuwin (HA01). Section 2 describes the signals associated with this event and the attempt made to localise it. Section 3 details an automatic detection algorithm that was developed to search for the same event in data from HA08S and was also used to characterise all events arriving at HA01 within an eight hour time period. HA08S and HA01 were the only operational CTBTO hydroacoustic stations in the Indian ocean at the time, The use of two of the separate events to check the calibration of the clock offset between RCS and HA01 is described in Section 4, while Section 5 details an initial attempt to estimate the probability that the signals at RCS and HA01 originated from the same source. Section 6 details the results of a search for the same event in data from HA08S, Section 7 provides a discussion that attempts to draw these various strands together, and the conclusions that can be drawn from this work are detailed in Section 8.

2 Signal descriptions and analysis

An underwater sound recorder operated by the Centre for Marine Science and Technology as part of IMOS and located at the seabed in 450 m water depth in the Perth Canyon (Rottnest Trench) received a transient signal at approximately 01:34:00 UTC on 8th March 2014. (This receiver is referred to as RCS in what follows.) After correction for clock drift the time of arrival of the peak energy in this signal has been calculated to be 01:33:44 UTC $\pm$ 4 seconds. The signal and its spectrogram are shown in Figure 1.

A search of data from the Comprehensive Test ban Treaty Organisation's hydroacoustic station off Cape Leeuwin, Western Australia (HA01) carried out by the authors has revealed a signal believed to be from the same source, with a peak energy arrival time of 01:34:50 UTC $\pm$ 3seconds. The signals and their spectrograms are shown in Figure 2.

Although the signal to noise ratio was poor on a single HA01 hydrophone, the signal at HA01 was strongly correlated between the station's three hydrophones which allowed the bearing of the source to be calculated with considerable accuracy. The result was a bearing of $301.6^\circ \pm 0.75^\circ$.

Projecting this bearing back from HA01 towards the source, computing the expected time difference between signals arriving at the two receiving stations as a function of distance along the track, and matching this to the measured time difference results in a most probable position for the sound source of 5.93°S , 77.22°E (see Figure 3). Calculations were carried out using Matlab's mapping toolbox and the WGS84 ellipsoid. Further details of this result are given in Table 1.

Uncertainties of $\pm 0.75^\circ$ in bearing from HA01 and ± 4 s in travel time difference result in the bounding box shown in Figure 3 and uncertainties of $\pm 1^\circ$ in bearing and ± 7 s in time difference result in the somewhat larger bounding box shown in Figure 4.

Calculated latitudes and longitudes of the vertices of the bounding box are given in Table 2. The bounding box is very elongated along the bearing from HA01 which is a result of the hyperbolic position line obtained from the arrival time difference being almost parallel to the bearing from HA01 at that range.

If the source was at the most probable position, then the signal would have been emitted at approximately 00:39:11 UTC.

The travel time differences have been calculated using acoustic group velocities estimated from numerical waveform modelling using an adiabatic normal mode model and climatological sound speed profiles for the appropriate time of year from the World Ocean Atlas 2005, (NOAA 2005). This gave a mean group velocity of 1486.20 m/s along the path from the most probable source location to HA01 and 1486.35 m/s along the path from the most probable source location to RCS.

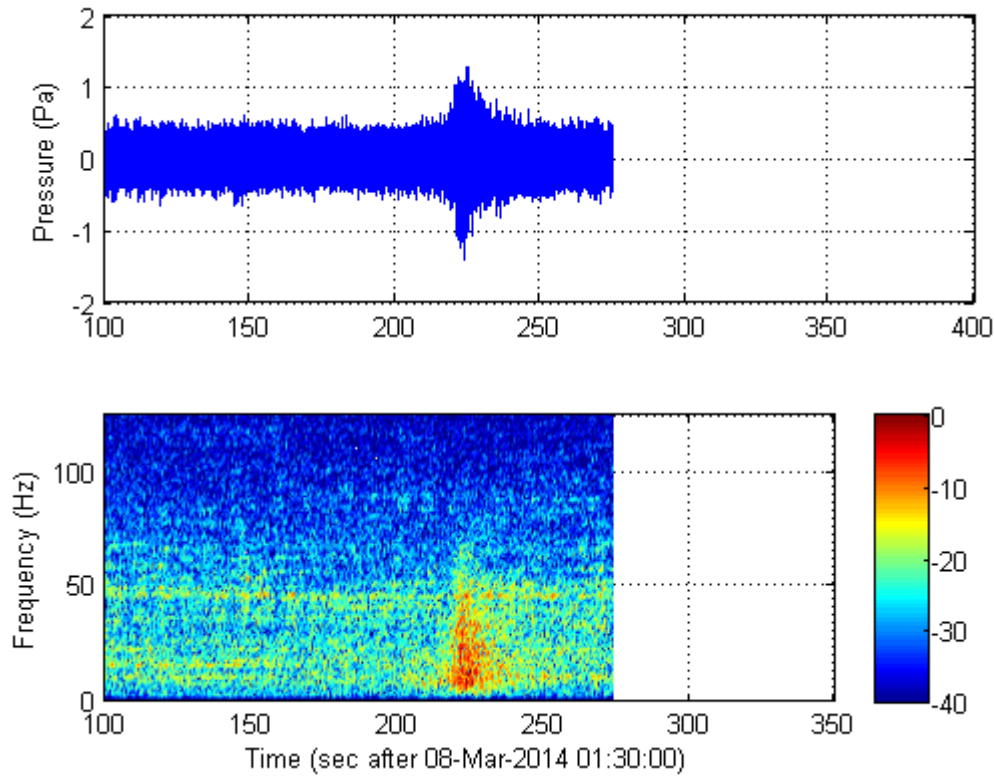


Figure 1. Time series (top) and spectrogram (bottom) showing the signal received at the Perth Canyon logger at 01:33:44 UTC on 8th March 2014. The spectrogram colour scale is in dB relative to the largest value in the spectrogram.

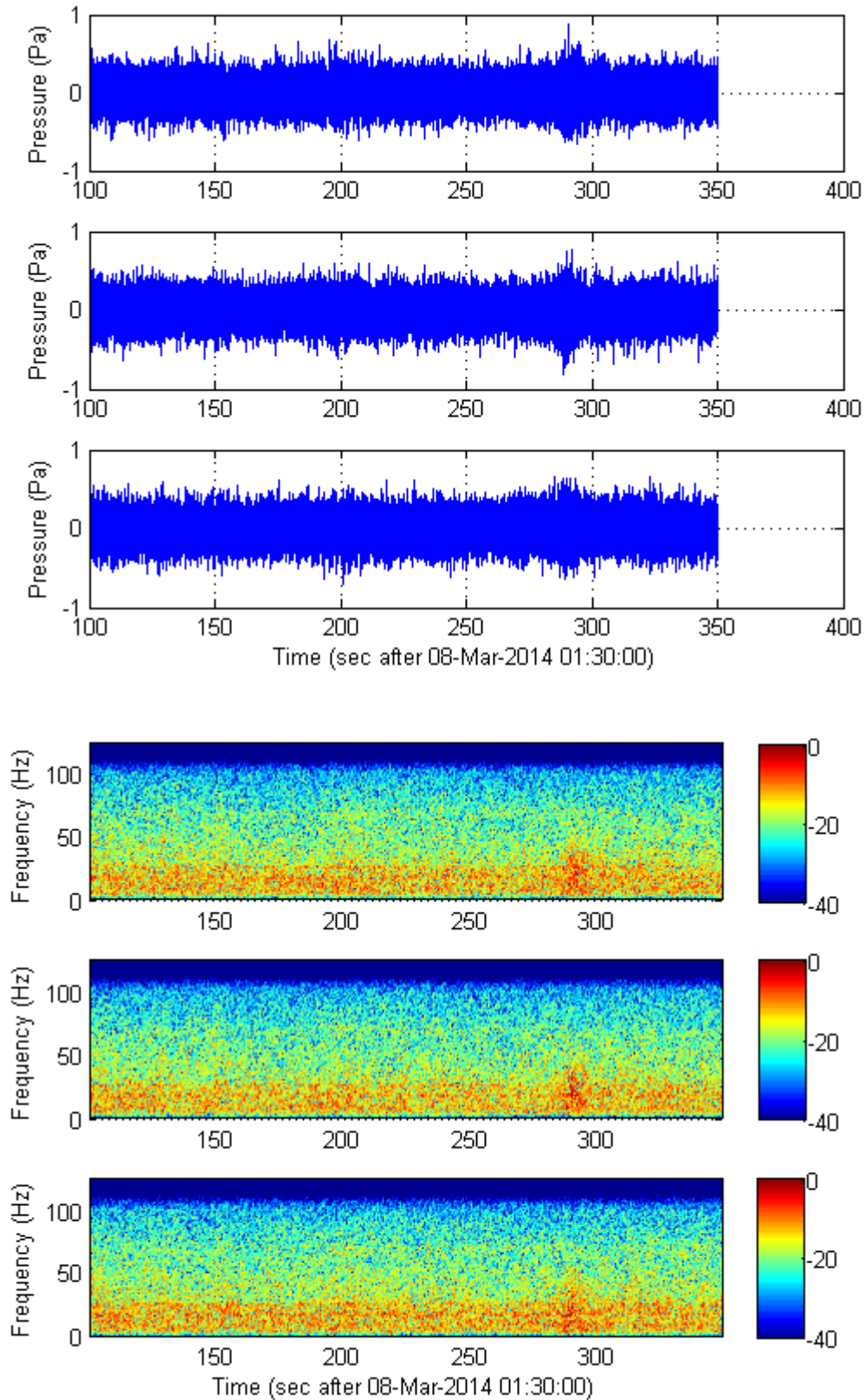


Figure 2. Time series (top) and spectrograms (bottom) showing the signals received at the three hydrophones of HA01 at 01:34:50 UTC on 8th March 2014. The spectrogram colour scales are in dB relative to the largest value in each spectrogram. Times scales are the same as for Figure 1.

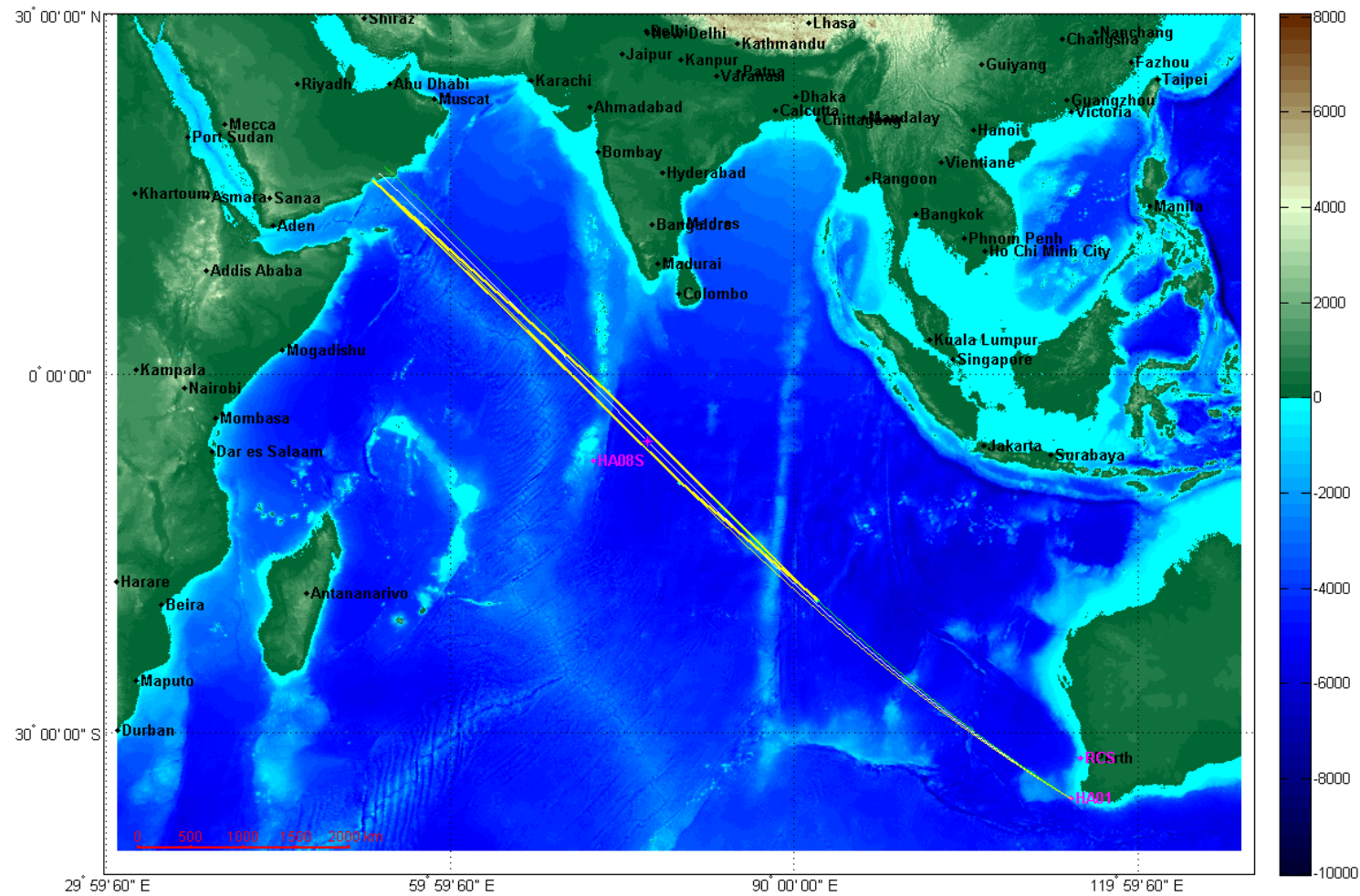


Figure 3. Map showing most probable location for the source of the received sound signals (magenta asterisk) and the uncertainty region (yellow polygon) based on an uncertainty of $\pm 0.75^\circ$ in the bearing from HA01 and a ± 4 s uncertainty in the difference between signal arrival times at RCS and HA01.

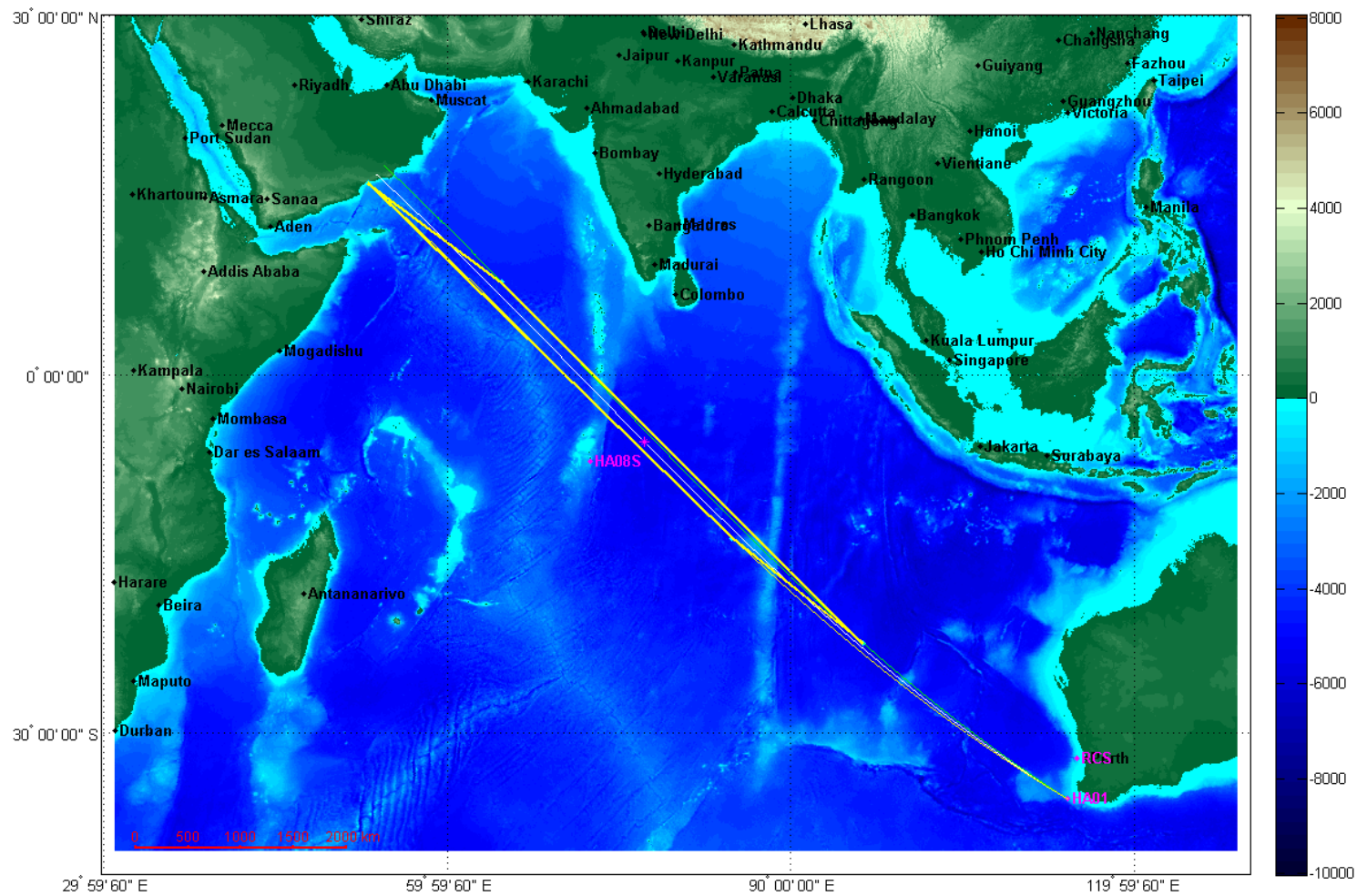


Figure 4. As for Figure 3 except the uncertainty region is based on $\pm 1^\circ$ in bearing and ± 7 s in arrival time difference.

Table 1.Details of most probable position.

Position	5.93°S, 77.22°E
Distance to HA01 (km)	4961.6
Distance to RCS (km)	4864.0
Signal travel time to HA01 (s)	3338.4
Signal travel time to RCS (s)	3272.4
Calculated arrival time difference (s)	66.0
Measured arrival time difference (s)	66.0

Table 2. Uncertainty box bounds

Vertex	Position with +/-0.75° bearing error and +/-4 s timing error	Position with +/-1° bearing error and +/-7 s timing error
Northwest	16.8°N, 53.3°E	16.6°N, 53.1°E
Northeast	1.6°S, 73.8°E	8.1°N, 64.7°E
Southwest	9.5°S, 80.1°E	14.2°S, 84.9°E
Southeast	19.5°S, 92.0°E	23.0°S, 96.3°E

3 Automated signal detection algorithm

3.1 General description

To provide an efficient means of searching for other arrivals a Matlab program was written that breaks the signals received at a CTBTO hydroacoustic station into short blocks and then iteratively calculates the bearing to any sources. A 95% block overlap was used so that any genuine signal should appear as a detection in multiple blocks.

Each block was processed by computing the bearing to the strongest source using the conventional plane-wave fitting method (Menke, 1984, del Pezzo and Giudicepietro, 2002), estimating that signal using a least squares process, subtracting the estimated signal from the received signals (with appropriate time adjustments) to produce modified signals, and then applying the plane wave fitting method to the modified signals. This process can be repeated an arbitrary number of times, but it was found that there was little point in proceeding beyond three iterations.

At each stage the integrity of the fitted bearing was checked by requiring a discrepancy in the arrival time differences of no more than 10 ms, a fitted group velocity in the range 1420m/s to 1550m/s, and at least one neighbouring detection with a bearing within $\pm 0.5^\circ$. The last criterion is based on the expectation that, because of the 95% block overlap, any genuine detection would occur in multiple adjacent blocks.

3.2 Mathematical derivation

Working in the horizontal (X, Y) plane, assume the incoming signal is made up of N plane waves coming from different directions. Then the received signal at time t at receiver $m \in \{1 \dots M\}$, located at x_m, y_m can be written:

$$p_1(x_m, y_m, t) = \sum_{n=1}^N s_n(t - \nu_{xn}x_m - \nu_{yn}y_m) + \varepsilon(x_m, y_m, t) \quad (1)$$

where

$\nu_{xn} = \frac{1}{c_{gn}} \cos \theta_n$ is the X-component of the slowness (inverse sound speed),

$\nu_{yn} = \frac{1}{c_{gn}} \sin \theta_n$ is the Y-component of the slowness,

c_{gn} is the plane wave group velocity,

θ_n is the direction in which the wave is travelling

$s_n(\cdot)$ is the waveform of signal n , and

$\varepsilon(x, y, t)$ is random noise that is assumed uncorrelated between receivers.

If the signals are ordered such that $|s_1| \gg |s_2| \gg |s_3|$ etc. then v_{x1} and v_{y1} can be estimated using the plane-wave fitting method (Menke, 1984, del Pezzo and Giudicepietro, 2002) and can therefore be considered known. We can write:

$$p_1(x_m, y_m, t) = s_1(t - v_{x1}x_m - v_{y1}y_m) + p_2(x_m, y_m, t) \quad (2)$$

$$\text{where } p_k(x_m, y_m, t) = \sum_{n=k}^N s_n(t - v_{xn}x_m - v_{yn}y_m) + \varepsilon(x_m, y_m, t).$$

So:

$$p_1(x_m, y_m, t + v_{x1}x_m + v_{y1}y_m) = s_1(t) + p_2(x_m, y_m, t + v_{x1}x_m + v_{y1}y_m). \quad (3)$$

Defining $t'_{m1} = t + v_{x1}x_m + v_{y1}y_m$ we can rewrite (3) as:

$$p_1(x_m, y_m, t'_{m1}) = s_1(t) + p_2(x_m, y_m, t'_{m1}). \quad (4)$$

If the $p_2(x_m, y_m, t'_{m1})$ are treated as noise that is uncorrelated between sensors then equation (4) defines a set of three equations in terms of the unknown $s_1(t)$. The assumption that the $p_2(x_m, y_m, t'_{m1})$ are uncorrelated between sensors is justified because, by definition, all signal components that are correlated between sensors when the time delays associated with t'_{m1} are applied are included in $s_1(t)$. The least squares estimate of $s_1(t)$ is simply the average over the appropriately delayed sensor outputs:

$$\hat{s}_1(t) = \frac{1}{3} \sum_{m=1}^M p_1(x_m, y_m, t'_{m1}) \quad (5)$$

Once $\hat{s}_1(t)$ has been estimated, the residual signals can be calculated using a rearrangement of (2):

$$p_2(x_m, y_m, t) = p_1(x_m, y_m, t) - \hat{s}_1(t - v_{x1}x_m - v_{y1}y_m) \quad (6)$$

The plane-wave fitting algorithm can then be applied to $p_2(x_m, y_m, t)$ and the process repeated to estimate $\hat{s}_2(t)$, and so on.

3.3 Results

The results of applying two iterations of this algorithm to the HA01 data from 00:10:00 UTC on the 8th March to 08:00:00 UTC on the same day are shown in Figure 1. These

results were obtained a block duration of 20s and a block overlap of 19s. A zero-phase high pass filter with a cut-off frequency of 3 Hz was applied to the data prior to processing to remove very low frequency flow noise.

The vast majority of the detections originated from the Antarctic sector, which extends from 158° to 209° , or from just outside it. The majority of these are likely to be ice cracking events, although some may be some seismic events along a seafloor spreading ridge between Australia and Antarctica.

The 01:34:50 arrival (indicated by the orange arrow) stands out as being the only signal over this time period that originated from a northerly direction.

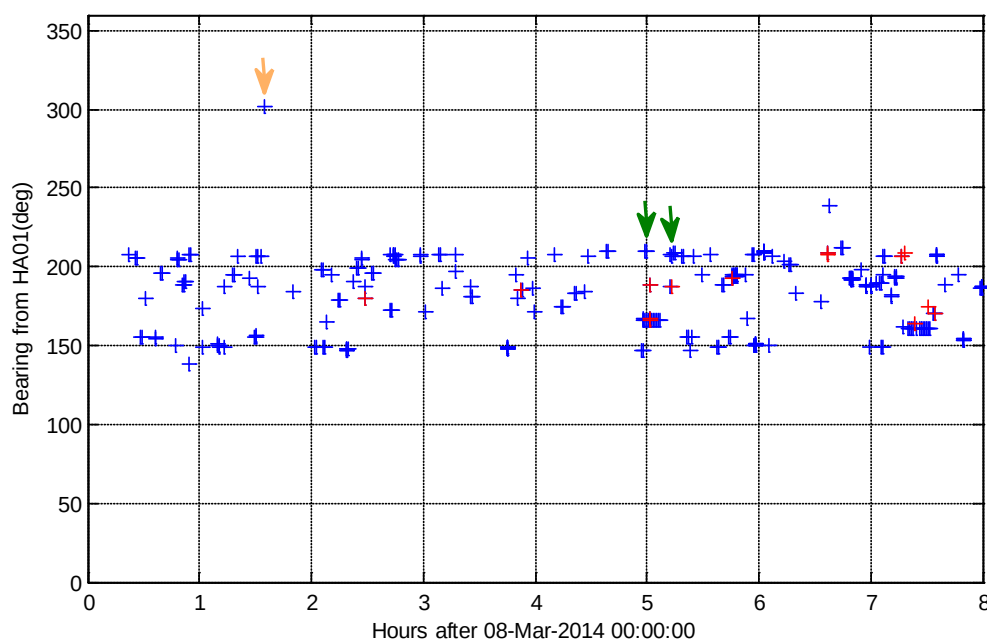


Figure 5. All automatic detections at HA01 between 00:00:10 and 08:00:00 UTC on 8th March 2014. The detection possibly associated with MH370 is indicated by the orange arrow. Blue crosses indicate detections made on the first iteration, red crosses are detections made on the second iteration. The dark green arrows indicate the two detections used for cross-checking the clock drift calibration.

4 Independent time calibration check

The estimate of the range of the source from HA01 is strongly dependent on the measured difference between the signal arrival times at RCS and HA01. The RCS recorder was deployed for a period of approximately three months during which its internal clock, which was set to UTC time prior to deployment and checked against UTC time after

deployment, gained 31.76 seconds (this is done in the noise logger hardware software using an external GPS and its UTC transmitted time). The time reference in both cases was the 1 pulse per second output from a GPS receiver. Assuming this drift rate was constant throughout the deployment, early on the 8th March 25.8 seconds must be subtracted from times according to the recorder clock in order to convert them to UTC.

An independent check of the accuracy of the drift correction of RCS compared with HA01 was carried out using two events other than the primary signal of interest, that resulted in signals that were received at both HA01 and RCS. The first of these resulted in a signal that was received at HA01 at 04:59:21 UTC on 8th March 2014 (17961 seconds after midnight) and came from a bearing of 210.2°. The corresponding signal arrived at RCS at 05:03:01 (18181 seconds after midnight). Both signals had high signal to noise ratio.

The second event was of much lower amplitude but came from a similar direction, and resulted in a signal that was received at HA01 at 05:14:28 UTC (18868 seconds after midnight) on the same day from a bearing of 208.5°. The corresponding signal was received at RCS at 05:18:13 (19093 seconds after midnight).

For each event, the arrival times and bearings at HA01 were used to compute a predicted arrival time at RCS, which was then compared to the measured arrival time. The results were computed using an assumed source range of 5000 km and an assumed group velocity of 1484 m/s and are given in Table 3.

Both arrivals came from a direction that is only about 15° from the direction of the line joining RCS and HA01, so the predicted arrival times at RCS depend only weakly on the source range assumption. A sensitivity study showed a 2 second increase in predicted arrival time if the assumed source range from HA01 was reduced to 1,000km, and a 0.2 second decrease if the source range was increased to 10,000km.

The assumed group velocity of 1484 m/s was obtained from numerical propagation modelling for a signal arriving from this direction and is considered accurate to within +/- 5 m/s. The effect of this uncertainty is considered along with other sources of error in the following analysis.

The uncertainties in the predicted arrival times at RCS were estimated based on a local plane geometry approximation to the ellipsoidal Earth in the vicinity of HA01 and RCS,

and the assumption that the source is an infinite distance away from the two stations. Note that this approximation has only been used to estimate the uncertainties. The actual arrival time calculations are based on ellipsoidal geometry and were carried out using Matlab's Mapping Toolbox.

With this approximation the arrival time at RCS is given by:

$$t_{RCS} = t_{HA01} + \frac{d \cos(\theta - \alpha)}{c_g}$$

where t_{HA01} is the arrival time at HA01 (seconds), d is the distance between HA01 and RCS (m), θ is the bearing of the source from HA01, α is the bearing of RCS from HA01 (13.65°), and c_g is the group velocity of the sound. The uncertainty in the predicted arrival time is given by:

$$\delta t_{RCS} = \left| \frac{\partial t_{RCS}}{\partial t_{HA01}} \right| \delta t_{HA01} + \left| \frac{\partial t_{RCS}}{\partial c_g} \right| \delta c_g + \left| \frac{\partial t_{RCS}}{\partial \theta} \right| \delta \theta$$

where $\delta \theta$ is the uncertainty in the bearing to the source (radians) etc. This leads to:

$$\delta t_{RCS} = \delta t_{HA01} + \Delta \frac{\delta c_g}{c_g} + \Delta \tan(\theta - \alpha) \delta \theta$$

where $\Delta = \left| \frac{d \cos(\theta - \alpha)}{c_g} \right|$ is the absolute value of the time delay between the signal

arriving at HA01 and at RCS in seconds.

In both cases the predicted and actual arrival times agree within uncertainties, with the maximum discrepancy being 3.2 seconds. This provides strong evidence that the clock synchronisation between HA01 and RCS is at least this accurate.

Table 3. Times are in seconds after 00:00:00 UTC on 8th March 2014.

Arrival time at HA01 (seconds)	Source bearing from HA01 (degrees)	Predicted arrival time difference (seconds)	Predicted arrival time at RCS (seconds)	Actual arrival time at RCS (seconds)	Predicted - actual (seconds)
17961 +/-2	210.2 +/-1	223.2 +/-2	18184.2 +/-4	18181 +/-2	3.2 +/-6
18868 +/-3	208.5 +/-1	225.0 +/-2	19093.0 +/-5	19093 +/-2	0 +/-7

5 Estimate of the probability that events at HA01 and RCS are from the same source

The estimate of the range of the source from HA01 depends on the assumption that the signals received at HA01 and RCS were generated by the same source and at the same time. An estimate has therefore been made of the probability that two signals arriving within a time window that would result in a valid fix, did so purely by chance. It should be noted that this estimate is based on very limited data and should therefore be treated as indicative only.

Let $P(A)$ be the *a-priori* probability that the HA01 and RCS arrivals are from the same source, and $P(B)$ be the *a-priori* probability that an arrival is observed at the RCS within a time interval of the arrival at HA01 that results in a valid fix.

Then $P(\bar{A}) = 1 - P(A)$ is the *a-priori* probability that the arrivals are from different sources and $P(\bar{B}) = 1 - P(B)$ is the *a-priori* probability that there was no arrival in the time interval required for a valid fix.

We want to calculate the *a-posteriori* probability that the arrivals are signals from the same source, given that the arrival at RCS occurred within the required time interval for a valid fix, i.e. $P(A | B)$, based on estimates of the probability of observing an arrival in the required time interval given that the signals are from the same source, i.e. $P(B | A)$, and

the probability of observing an arrival in the required time interval given that the signals are not from the same source, $P(B | \bar{A})$.

This can be done using Bayes Theorem (Walpole & Myers, 1985) :

$$P(A | B) = \frac{P(A)P(B | A)}{P(B)} \quad (7)$$

which can also be written:

$$P(A | B) = \frac{P(A)P(B | A)}{P(A)P(B | A) + P(\bar{A})P(B | \bar{A})} \quad (8)$$

So the problem reduces to estimating the various probabilities on the right hand side of Eq. (8).

If the signals are from the same source, then the RCS arrival must occur within the valid time interval, so $P(B | A) = 1$.

$P(B | \bar{A})$ can be estimated by looking at the statistics of similar arrivals at RCS over a long period of time. So far we have only looked at five hours of data, which is only enough for a very rough estimate, however even this is potentially useful as a starting point. There were two arrivals at RCS in that five hour period with broadly similar characteristics (duration and bandwidth) to the HA01 arrival – the one within the feasible time interval and another about 3 hours later. (The second arrival was actually a bit different in that it had a different pulse shape and a somewhat broader bandwidth, but for the purposes of this analysis we'll consider them as two events from the same process.) If we assume these pulses originate from a Poisson process, then an estimate of the process's rate parameter is $\lambda = 2/(5 \times 3600) = 1.11 \times 10^{-4}$ events per second. The probability of receiving k arrivals within a given time interval T is then given by the Poisson probability mass function:

$$P(X = k) = \frac{(\lambda T)^k}{k!} e^{-\lambda T} \quad (9)$$

The probability of receiving one arrival within the feasible time interval is then:

$$P(B | \bar{A}) = P(X = 1) = \lambda T e^{-\lambda T} \quad (10)$$

where T is the duration of the feasible time interval and can be determined from the geometry. Valid arrival time differences range from -200 s (for a source 40km from HA01, negative indicates the signal arrives at HA01 first) to +80 s for a source at 15,000km range, giving an interval duration of $T=280$ s. Substituting this into Eq. (10) with the previously calculated value of λ gives $P(B | \bar{A}) = 0.030$.

The other term required in order to evaluate Equation (8) is $P(A)$, the *a-priori* probability that the HA01 and RCS arrivals are from the same source. This can be interpreted to mean the probability that one would assign given only information about the characteristics of the two signals and without any knowledge of their relative arrival times. The arrivals at HA01 and RCS have very similar durations and spectra that are unlike the majority of other signals received at HA01, and the ratio of their amplitudes is consistent with model predictions. $P(A)$ should therefore be reasonably high, however as it is unclear (at least to the author) how this could be determined from the available data, results are plotted below in Figure 1 with $P(A)$ as a parameter.

It can be seen that providing $P(A)$ is greater than 0.03, $P(A | B)$ will be greater than 0.5. Furthermore, if $P(A) > 0.5$, then $P(A | B) > 0.97$.

Although the dependence of the final result on the difficult to determine *a-priori* probability, and the small amount of data used to estimate $P(B | \bar{A})$ make this analysis somewhat unsatisfactory, it does indicate that it is quite likely that the signals received at HA01 and RCS are from the same source. Analysis of considerably more data from both HA01 and RCS would provide a more robust result, but this has not been attempted.

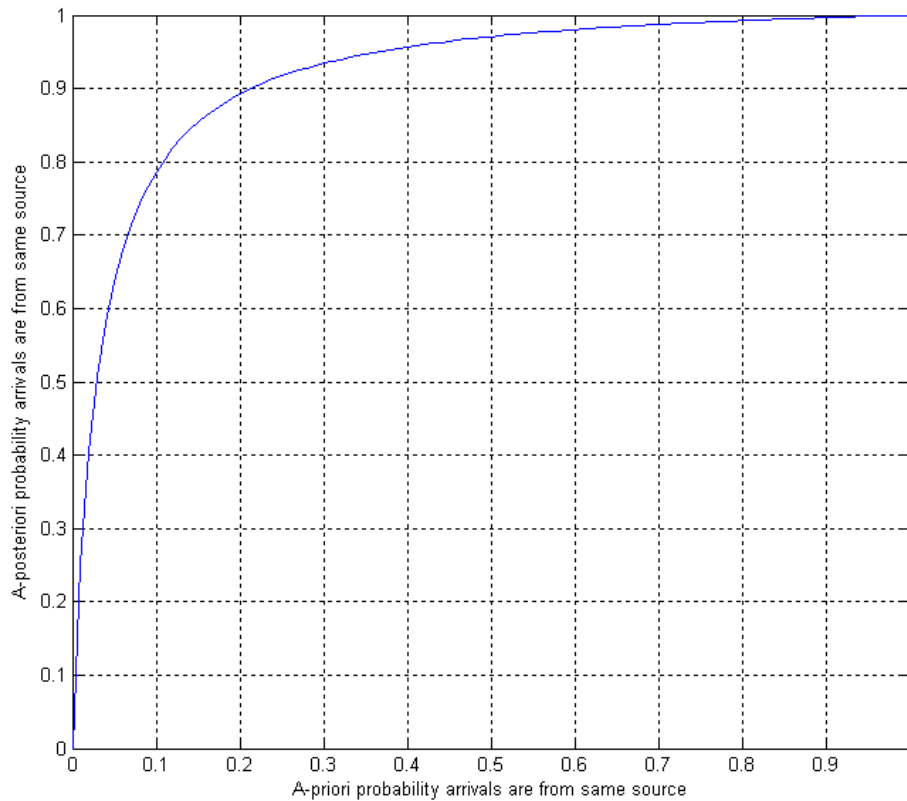


Figure 6. Plot of the a-posteriori probability against the a-priori probability that the arrivals at the two stations are from the same source.

6 Analysis of data from HA08S

Figure 6 shows the automatically calculated arrival bearings as a function of arrival time for all detections at HA08S. These results are summarised in Table 1. The two lines of detections that occur at almost constant bearings of 35° and 113° have characteristics consistent with offshore seismic surveys, which appear to have been taking place off Sri Lanka or in the Bay of Bengal, and off the northwest coast of Australia respectively. There were two other detections, one at about 700 seconds from a bearing of 168.1° and one at 7135 seconds from a bearing of 93° . Position lines corresponding to these various detections are shown in Figure 8.

A curve showing the expected arrival bearing at HA08S as a function of arrival time for the signal detected at HA01 at 01:34:50 on 8/3/2014 UTC is overlaid on Figure 6. It can be seen that neither of the discrete detections is consistent with the HA01 data.

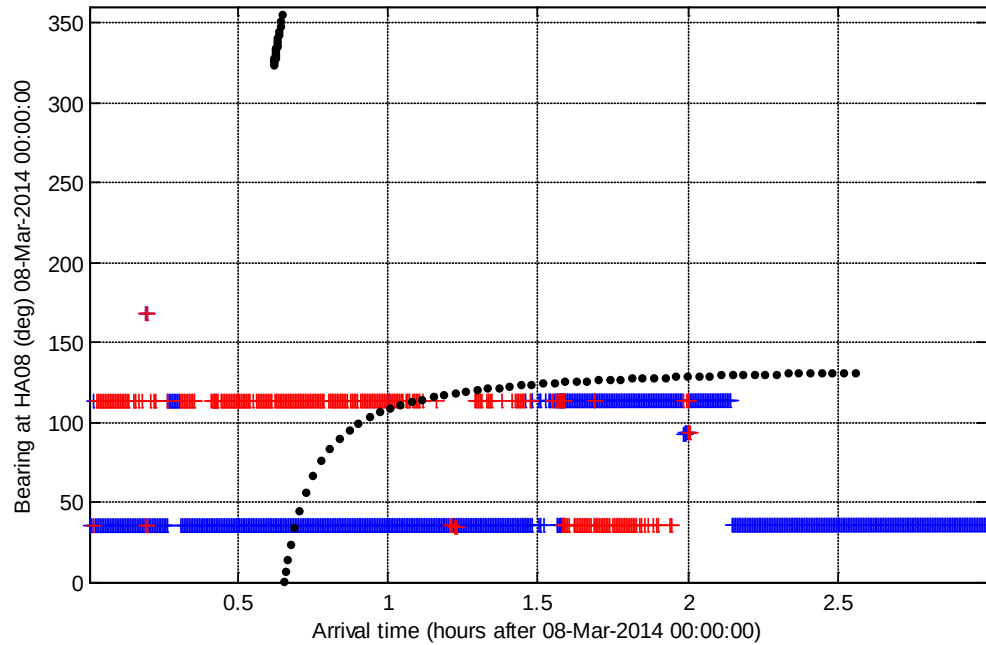


Figure 7. All automatic detections at HA08S between 00:00:00 and 03:00:00 UTC on 8th March 2014. The black dotted line shows the expected arrival bearing at HA08S as a function of arrival time for the signal detected at HA01 at 01:34:50 on 8/3/2014 UTC.

Table 4 Summary of HA08S detections

Time range (hours after 00:00:00 on 8/3/2014 UTC))	Bearing range (deg)	Figure 7 line colour	Likely source
0 to 3.00	34.9 - 35.4	Yellow	Seismic survey off Sri Lanka or in Bay of Bengal
0 to 2.14	113.0 - 113.6	Green	Seismic survey off NW Australia
0.19 to 0.20	168.05	White	Unknown. Possibly ice cracking noise or a small seismic event.
1.98 to 2.00	92.5 to 93.5	Red	Small earthquake in the Java Trench

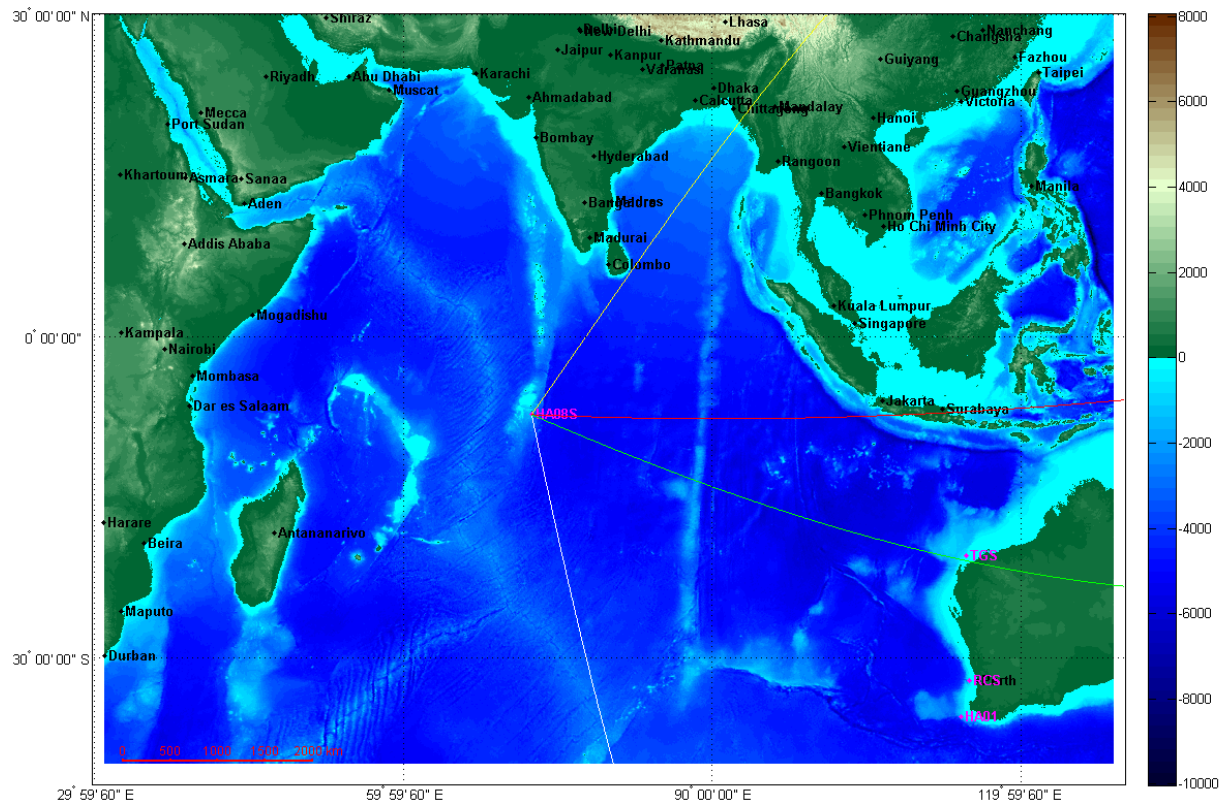


Figure 8. Map showing position lines for the HA08S detections listed in Figure 4. The location of the TGS seismic survey is shown by the magenta dot labelled "TGS".

Enquiries with industry contacts revealed that at the time MH370 is thought to have crashed, two seismic surveys were taking place off the northwest coast of Australia, close to the position line based on the 113° bearing from HA08S. These were the Woodside Centaurus 3D Survey and the TGS Huzzas 3D Survey. Both operators have provided the navigation data from the relevant survey lines, which are plotted in Figure 8.

The Woodside survey line was over a fairly flat seabed in just over 1000 m of water, whereas the TGS survey was over the continental slope in about 260 m of water. The time of reception of the last shot and the change in bearing with time both clearly indicate that the signals received at HA08S were from the TGS survey. The Woodside survey was not detected. This is consistent with the physics of acoustic propagation which requires a shallow sound source, such as a seismic survey airgun array, to be over a favourably sloping seabed in order to efficiently couple the sound into the Deep Sound Channel and achieve very long range propagation.

The navigation data provided by TGS allowed a check to be made of the bearing accuracy of HA08S. The results are plotted in Figure 9 and show a mean bearing offset of 0.6° . The resolution of the bearing measurement is sufficient to clearly track the northeast to southwest motion of the survey vessel.

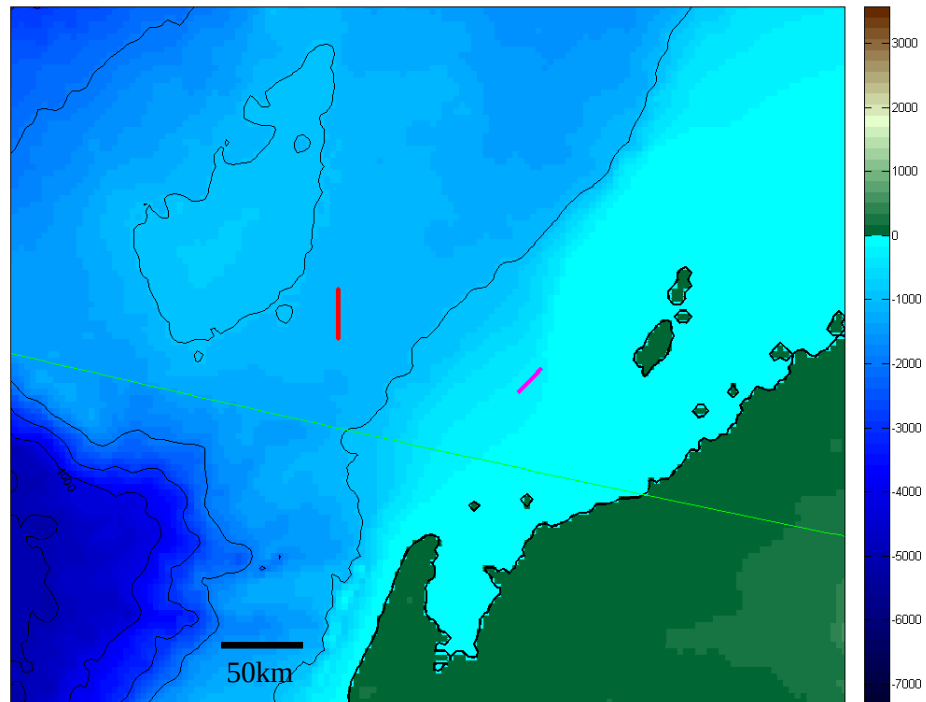


Figure 9. Map of Northwest Cape region, Australia, showing seismic survey lines being shot by Woodside (red) and TGS (magenta) around the time MH370 is thought to have crashed. The green line is the position line from HA08S

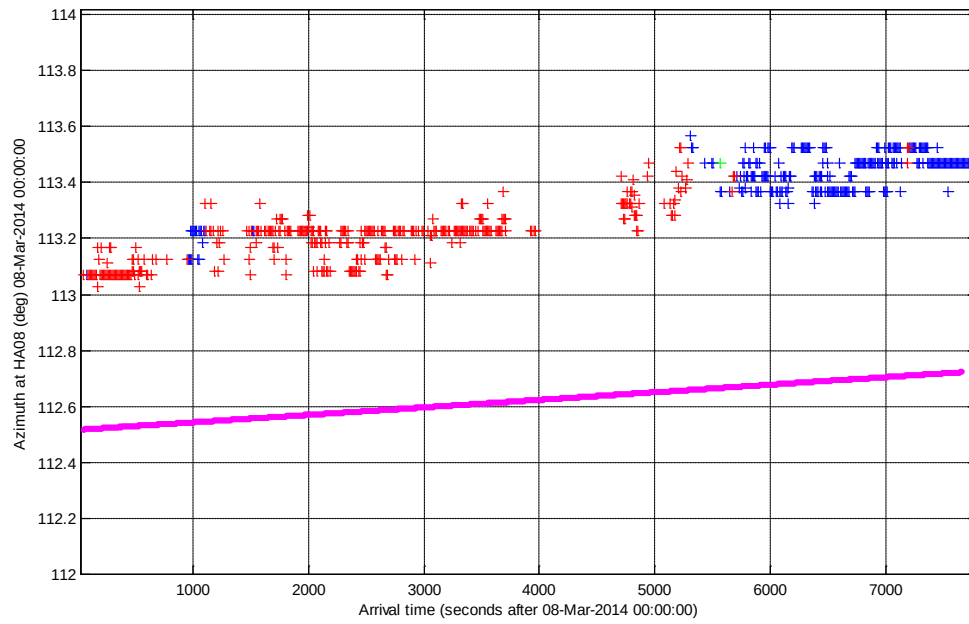


Figure 10. Plot of bearing vs. time for detections at HA08S that correspond to the TGS Huzzas 3D seismic survey (crosses). The magenta line is the calculated bearing based on the survey navigation data provided by TGS.

7 Discussion

Even with the largest likely uncertainties of $\pm 1^\circ$ in the bearing from HA01 and ± 7 s in the difference between the times of arrival of the signals at HA01 and RCS, the uncertainty region for the source location does not intersect the arc defined by the last satellite handshake (see Figure 11). An investigation of the timing error required to place the source location on the satellite handshake arc indicated that the arrival time difference would have to be 17 seconds less than measured, which is well outside uncertainties. Our estimate of the time of arrival of the signal at HA01 has been confirmed by Mark Prior from the CTBTO in Vienna, and the correction for clock drift in the RCS recorder has been checked and rechecked. Furthermore, the time calibration checks described in Section 4 provide independent confirmation that time of arrival measurements made at the two stations are consistent once the clock drift correction has been applied.

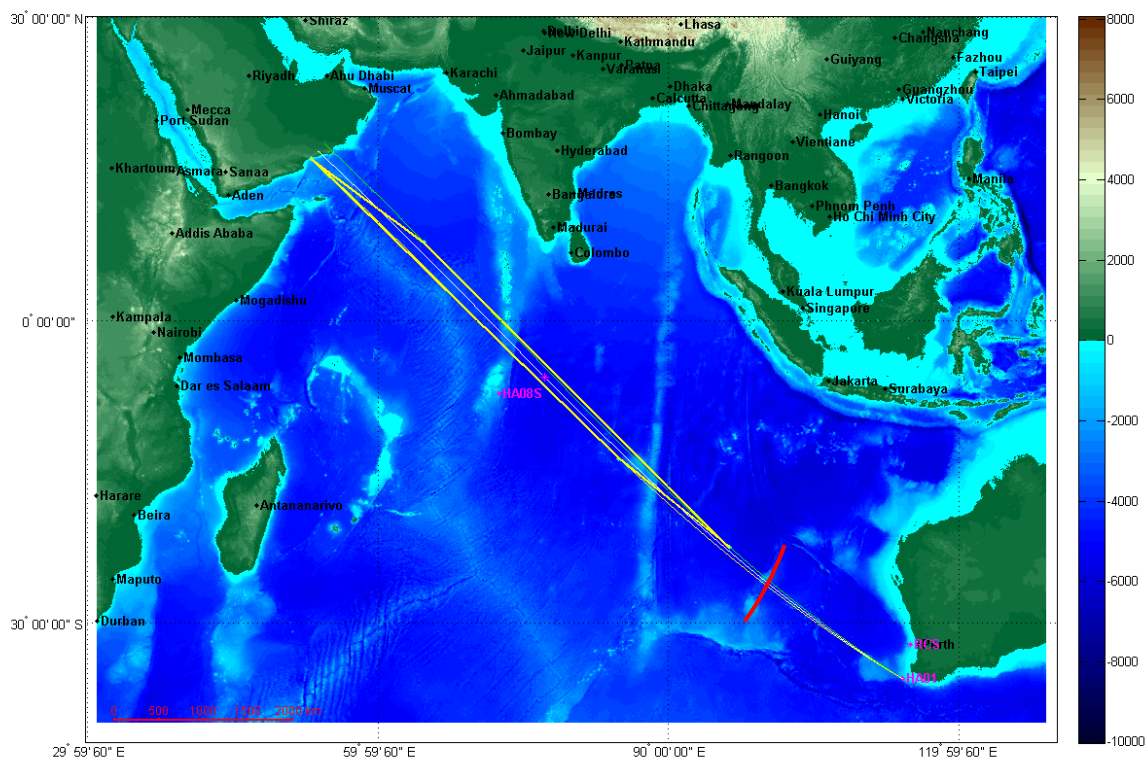


Figure 11. Map showing the uncertainty region (yellow polygon) based on an uncertainty of $\pm 1^\circ$ in the bearing from HA01 and ± 7 s in the arrival time difference, with the approximate location of the arc defined by the satellite handshake data superimposed (red line).

It is therefore apparent that the acoustic measurements are inconsistent with the satellite handshake data. We consider three possible explanations for this:

1. The signals received at HA01 and RCS are from the same acoustic event, but the source of the signals is unrelated to MH370.

This seems the most likely explanation as the characteristics of the received signals are similar to those of some natural events. The origin of these events is not certain, but they are thought to be of natural seismic origin. Events from this particular direction are not particularly common, but do occur. An analysis of five years of historical data from HA01 that was available to the authors indicated that over that period there was an average of about one event per day from this direction. However, on closer inspection it became apparent that most of these events occurred on one day, and then became progressively less frequent. (This is a common pattern with natural seismic events.) It would be useful to analyse data from the year or so leading up to 8th March 2014 in order to obtain a better estimate of the statistics relevant to that date, but this has not been done.

2. The signals received at HA01 and RCS are from different acoustic events, which may or may not be related to MH370.

If this were the case then the distance determination would be invalid and the source of the signal received at HA01 could be anywhere along the position line defined by the measured bearing from HA01, including on the satellite handshake arc. However, the analysis carried out in Section 5, while imperfect, indicates that it is more likely that the signals received at the two stations are from the same event. Also, if the sound source was on the satellite arc the signal would have been generated at 01:16:19 UTC on 8th March, some 25 minutes after the latest time it is thought that MH370 could have hit the water (00:51:00 UTC according to ATSB). The signal could, however, have been generated by the implosion of sinking debris if this occurred well after the initial impact, although this would be expected to be within a few km of the surface impact site.

3. The signals received at HA01 and RCS are from the same acoustic event, and the source of the signals is related to MH370, but there is a problem with the position line determined from the satellite handshake data.

If this were the case then the location of the aircraft would be much further from the Australian coast than the satellite handshake data would indicate. This explanation seems unlikely given the intense scrutiny that the satellite handshake data has been subjected to, however it is worth noting that the time at which the signal was received at HA01 is consistent with the time the aircraft is thought to have crashed if the impact occurred in the central portion of the uncertainty box plotted in Figure 3. To make this explicit, Figure 12 plots the source locations for a signal generated at the earliest possible impact time (00:19:49 UTC - north-western triangle), at the latest possible impact time (00:51:00 UTC - south-eastern triangle) and following the arc given by the signal received at HA01, based on the computed acoustic travel time to HA01 assuming a mean group velocity of 1486 m/s. These extreme locations can be seen to bracket the most probable position obtained solely from the acoustic data.

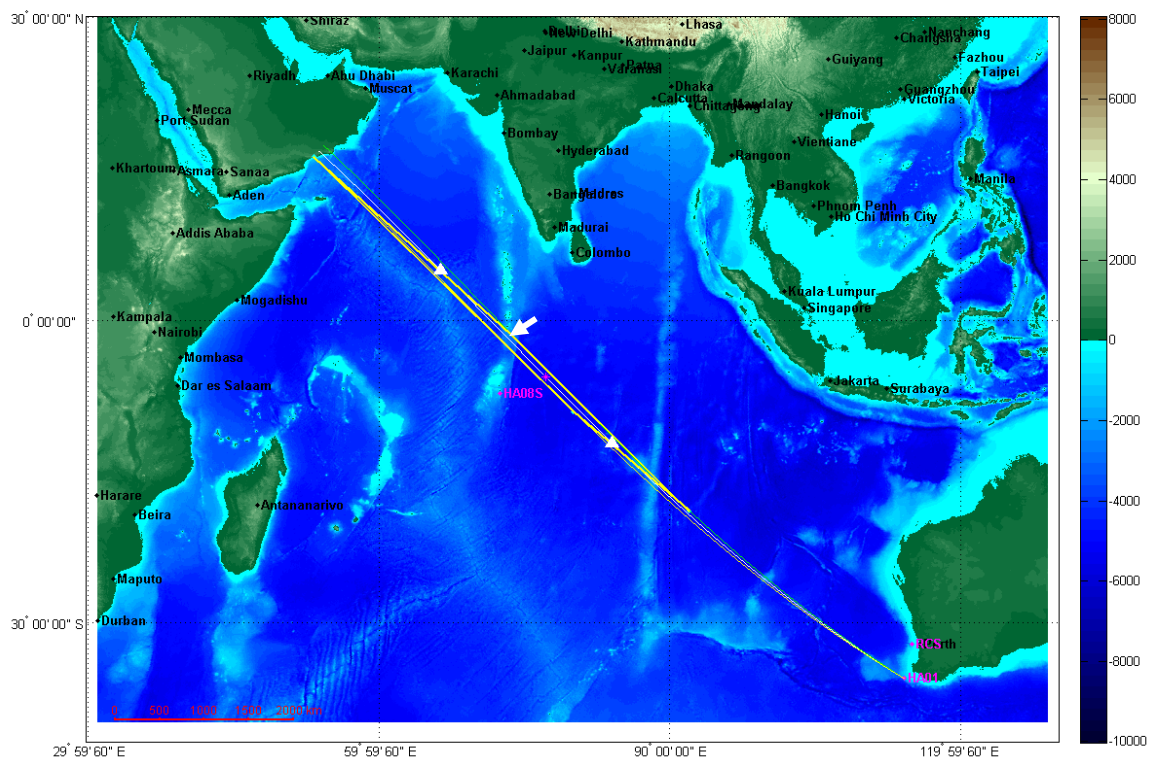


Figure 12. Map showing the smaller uncertainty box with range limits based on the earliest and latest times it is thought the aircraft could have hit the water superimposed (white triangles), assuming the aircraft crash was along the arc from HA01 given by the signal of interest. The arrow indicates the only area between these range limits that also satisfies the requirements for efficient coupling of sound from a near-surface source into the Deep Sound Channel.

Another consideration is that for sound from a near-surface sound source to be detectable at ranges of thousands of kilometres it must couple into the Deep Sound Channel well. A near surface source will generate sound in the water mostly at steep angles and will not efficiently couple into the Deep Sound Channel unless the seabed bathymetry is favourable. For the signal sound energy to couple into the Deep Sound Channel it is necessary for the seafloor at the source location to be sloping downwards towards the receiver. This allows sound reflected from the seabed to be directed into the Deep Sound Channel where it can propagate with very little attenuation. This process is most effective when the water depth is also shallow enough that the seabed intersects the Deep Sound Channel, the axis of which lies somewhere between 800-1200 m depth depending on latitude.

The only area between the locations defined by the earliest and latest impact times along the arc from HA01 given by the signal of interest that satisfies this criterion is on the Chagos-Laccadive Ridge, the submarine ridge that extends northwards from HA08S. The bearing from HA01 crosses this ridge at approximately 2.3°S, 73.7°E, just south of the Maldives, as indicated by the arrow in Figure 12.

The water depths along this bearing of 301.6°, and bearings 0.75° either side of it are plotted as a function of range from HA01 in Figure, along with plots of the computed acoustic transmission loss between a 20 Hz sound source, 30 m below the water surface, and HA01. Note that the way the transmission loss is plotted, an upward excursion corresponds to a reduction in transmission loss, which corresponds to an increase in the received signal. The transmission loss was computed using the Parabolic Equation acoustic propagation model RamGeo (Collins 1993), using range-dependent sound speed profiles from the World Ocean Atlas, 2005 (NOAA 2005), and assuming a low reflectivity, silt seabed with geoacoustic parameters from Jensen (2011). The effect would be enhanced if the seabed were harder and more reflective. Transmission loss calculations were also carried out for frequencies of 10 Hz and 40 Hz and produced similar results.

As expected, oceanographic features that result in sharp reductions in water depth give rise to corresponding reductions in transmission loss, which would correspond to an improved likelihood of detecting a signal for a source at that location. The effect does, however, depend critically on water depth. It can be seen that the Chagos-

Laccadive Ridge is too deep to have much effect along a bearing of 301.6° where it reaches a minimum depth of about 2200 m, but reduces the transmission loss by 20 dB along 302.36° where it rises to a minimum depth of just over 1000m.

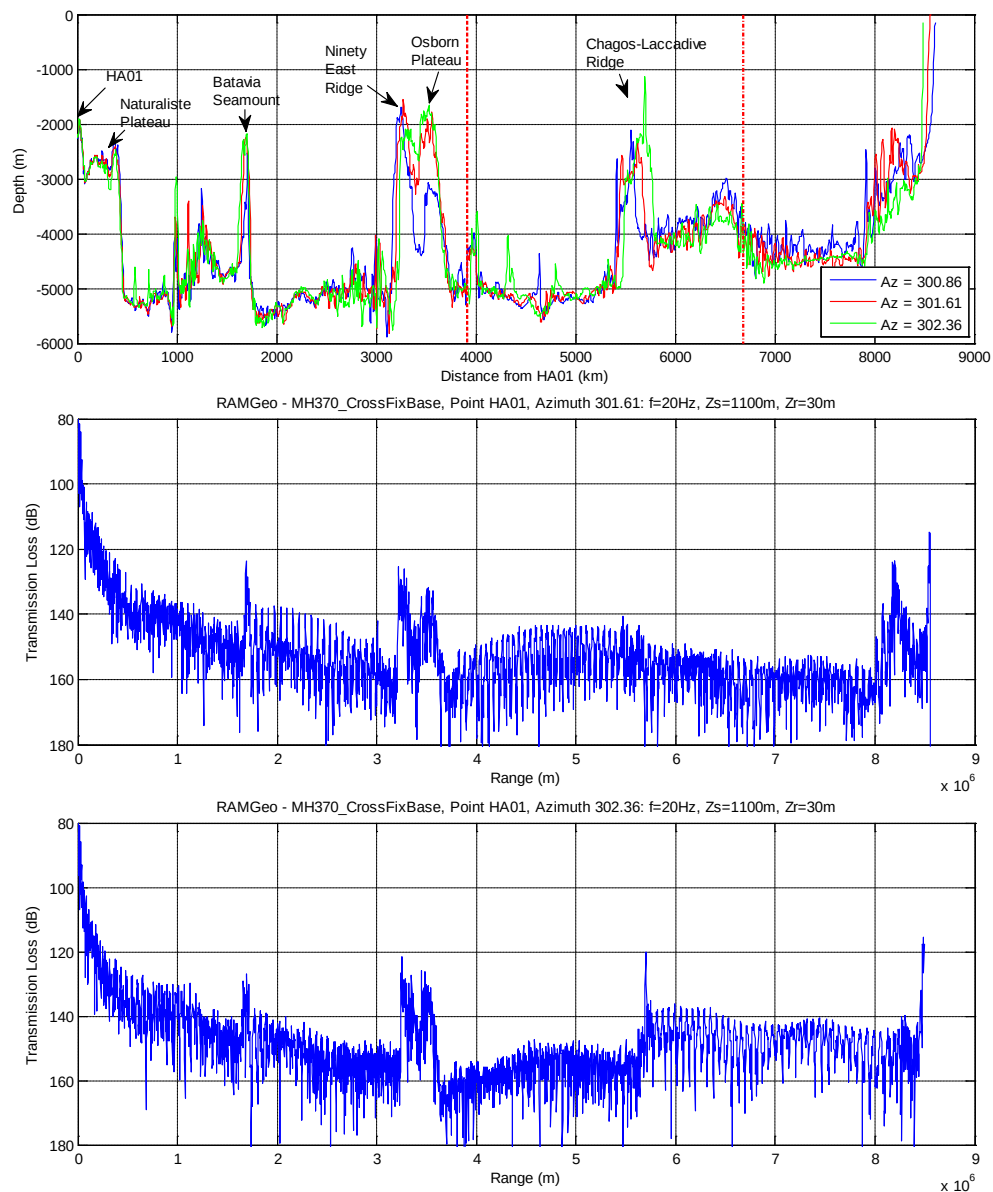


Figure 13. Top panel shows the bathymetry along the position line defined by a bearing of 301.61° from HA01, and bearings 0.75° either side of this as a function of distance from HA01. Red vertical lines indicate the range limits computed from the earliest and latest possible impact times. Various oceanographic features are indicated. The middle and lower panels show the computed transmission loss between a 20Hz acoustic source, 30m below the water surface, and HA01 as a function of range from HA01. Middle panel is for a bearing of 301.61° , lower panel is for 302.36° .

The above argument applies only if the source of the sound was the impact of the plane on the water surface. If it was the implosion of sinking debris, some time after

the impact, then the location could be closer to HA01 than the range limits based on the impact time would indicate. If the implosion occurred in the upper one hundred metres or so of the water column then the same considerations regarding bathymetry would apply, and other possible locations for the aircraft would include the Ninety East Ridge and the Osborn Plateau. However, if an implosion occurred deeper, then efficient deep sound channel propagation would be possible irrespective of the bathymetry. This effect is illustrated in Figure 14, which plots the transmission loss as a function of the distance of the source from HA01, and the source depth.

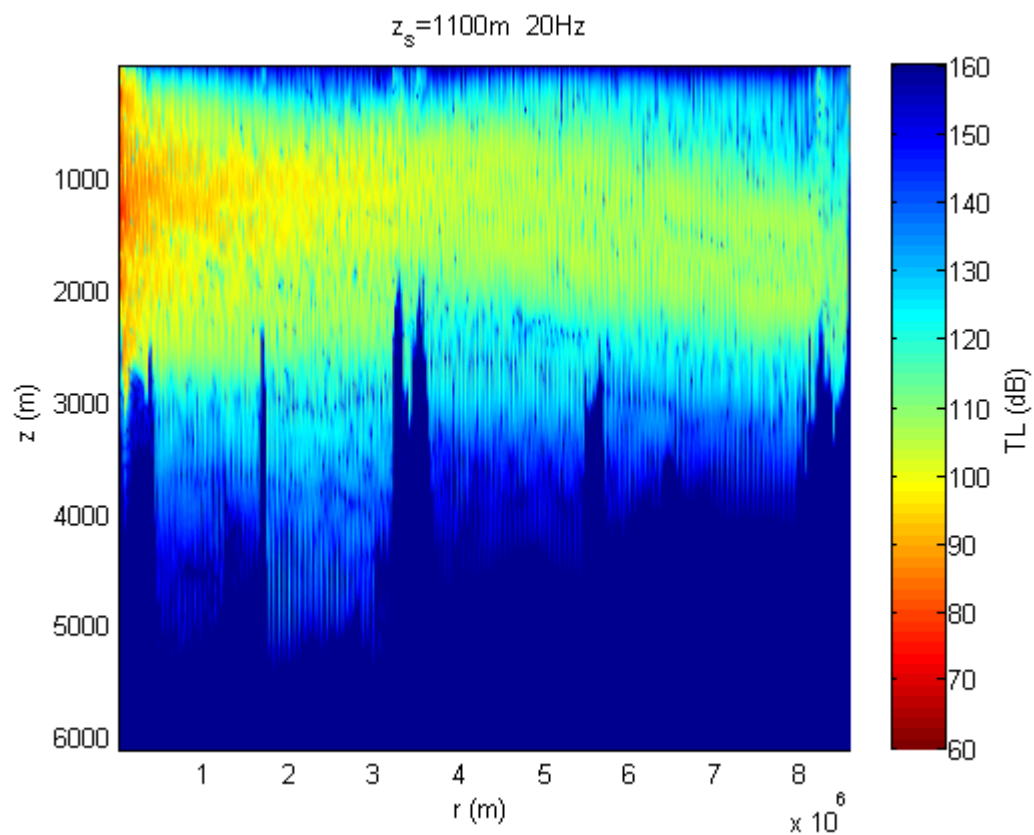


Figure 14. Transmission loss for 20 Hz acoustic signals arriving at HA01 at a bearing of 301.61° as a function of the distance of the source from HA01 and the depth of the source.

CMST are attempting to obtain other sources of acoustic data from the Indian Ocean which may contain the signals discussed here and may allow a better definition of the acoustic data discussed here. These sources are:

1. IMOS passive acoustic recorder located west of Dampier, North Western Australia (19° 23.06' S, 115° 54.92). This mooring is expected to be recovered in early August 2014. This receiver is on a 33% duty cycle.
2. IMOS passive acoustic recorder located north of Broome, North Western Australia (15° 29.002' S, 121° 15.06' E). This mooring is expected to be recovered in early August 2014. This receiver is on a 33% duty cycle.
3. Marine seismic survey recordings operating in the northern Indian Ocean over the time frame of aircraft impact. We have had an excellent response from the marine seismic community with regard to searching for seismic surveys operating in this area over the time frame of interest. If we can locate seismic surveys which were operating over the time frame of interest in areas where it is possible they may have received the signals discussed here we will request their streamer data to check for signals of interest. Marine seismic surveys typically collect data at a 60-80% duty cycle although analysis would be complicated by the presence of the intense marine seismic source signals.

8 Conclusions

Analysis of the low frequency acoustic signals that were received at RCS and HA01 around the time that MH370 disappeared has identified one acoustic event that could possibly be linked to the loss of the aircraft. It is also possible that this was a natural event and unrelated to the aircraft.

A detailed analysis of these signals has resulted in an approximate localisation for the source of the signals that is compatible with the time of the last satellite handshake with the aircraft, but not compatible with the satellite to aircraft range derived from this handshake which is being used to guide the search. There appear to be three possible explanations for this discrepancy:

1. The signals received at HA01 and RCS are from the same acoustic event, but the source of the signals is unrelated to MH370.
2. The signals received at HA01 and RCS are from different acoustic events, which may or may not be related to MH370.

3. The signals received at HA01 and RCS are from the same acoustic event, and the source of the signals is related to MH370, but there is a problem with the position line determined from the satellite handshake data.

Of these, the first explanation seems the most likely as the characteristics of the signals are not unusual, it is only their arrival time and to some extent the direction from which they came that make them of interest.

If the second explanation was correct then there would still be some prospect that the signal received at HA01 could be related to the aircraft, in which case the combination of the HA01 bearing and the position arc derived from the satellite handshake data would provide an accurate location on which to base a search. However, the analysis carried out here indicates that, while not impossible, this explanation is unlikely.

The third explanation also seems unlikely because of the intense scrutiny the satellite handshake data has been subjected to, however, should the arc defined by the handshake data be called into question, the various acoustic considerations discussed here would suggest that a reasonable place to look for the aircraft would be near where the position line defined by a bearing of 301.6° from HA01 crosses the Chagos-Laccadive Ridge, at approximately 2.3°S , 73.7°E . If the source of the detected signals was the aircraft impacting the sea surface then this would most likely have occurred in water depths less than 2000 m and where the seabed slopes downwards towards the east or southeast. This consideration could be used to further refine the search area. If, instead, the received sounds were due to debris imploding at depth it is much less certain where along the position line from HA01 this would have occurred.

The CMST are still to receive further passive acoustic data which may improve the location estimate provided here and allow a better definition of the signal source.

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